



Generalizability of Self-supervised Training Models for Digital Pathology: A Cross-country Comparison in Colorectal Cancer

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ABSTRACT

The high-resolution and wide field-of-view characteristics of whole slide images (WSIs) often make refined annotation impossible. Fortunately, self-supervised learning (SSL) provides an ability to train models on large-scale unlabeled WSIs directly. However, considering the difference of ethnic and diagnostic criteria, the self-supervised model trained over the data of only one country will inevitably face the problem of poor generalization. In this paper, for colorectal cancer (CRC), we demonstrate the superiority and generalizability of multi-country self-supervised training in four CRC-related medical tasks. We build a multi-country CRC dataset with 2,249 WSIs from Australia and the United States for SSL. The pre-trained model by SSL verifies the superiority over four multi-country downstream tasks. Our results show that the performance over the multi-country dataset is better than that over the ImageNet pre-trained model in most downstream tasks. Furthermore, better generalization can also be achieved in the external datasets.

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1. Introduction

Histopathological diagnosis is the gold standard for clinical tumor diagnosis and prognosis evaluation (Srinidhi et al., 2021; Baxi et al., 2021). With the advent of pathological scanners, pathological diagnosis has entered the digitization stage, and the emergence of a large number of whole-slide images (WSIs) has prompted the continuous development of artificial intelligence (AI) in this field. Studies (Campanella et al., 2019; Yu et al., 2021; Lu et al., 2021; Saldanha et al., 2022) have shown that models trained over a large number of WSIs can achieve comparable results with human pathologists on a range of common cancers, such as colorectal cancer (CRC).

Globally, CRC is the second and third most frequently diagnosed cancer in females and males, respectively. Furthermore,

the incidence and mortality of CRC vary widely worldwide. (Islami et al., 2021; Arnold et al., 2017). Deep learning technology has enabled a series of breakthroughs in tumor and normal classification of CRC (Zhou et al., 2020; Yu et al., 2021), prediction of microsatellite instability (MSI) (Kather et al., 2019b; Pollett, 2020; Yamashita et al., 2021), CRC grading (Shaban et al., 2020), and prognosis analysis (Kather et al., 2019a; Skrede et al., 2020).

Training from scratch requires large-scale labeled data, which is unrealistic in the histopathology scenario due to the large number of pixels per WSI and the level of professional knowledge required for labeling (Srinidhi et al., 2021). To alleviate this problem, ImageNet weights pre-trained on natural images are often used for model initialization. Another solution is the self-supervised learning (SSL) method (Chen et al., 2020; Chen and He, 2021; He et al., 2021; Xie et al., 2021), which can be employed for unlabeled data. The self-supervised method learns the similarity and dissimilarity of the images un-

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der different views through data augmentation. Current self-supervised methods have exhibited a superior ability to significantly facilitate the learning of downstream tasks in pan-cancer (Ciga et al., 2022; Wang et al., 2021), breast cancer (Chen and Krishnan, 2022; Srinidhi et al., 2022), and lung cancer (Li et al., 2021).

Notably, another inevitable problem in the histopathology scenario is the difference in tissue preparation and image scanning quality in various hospitals. Therefore, studies in one country or center may lead to overly optimistic conclusions (Van der Laak et al., 2021; Hashimoto et al., 2020). Because the algorithms developed require robust adaption to various changes in clinical practice, multi-country research is considered to be urgently needed. Some studies have adopted multi-country test datasets to verify the superiority and generalization of the algorithm (Naik et al., 2020; Lu et al., 2021; Laleh et al., 2022). However, as one of the most prevalent cancer types worldwide, CRC-specified self-supervised work in multiple countries has not yet been discussed in detail.

This study explores SSL on multi-country data for CRC and evaluate four downstream tasks. At the SSL pre-training stage, we employed the SimSiam training method over multi-country datasets, which included two large-scale CRC datasets: the MCO dataset from Australia and the TCGA dataset from the United States. In the downstream validation stage, the pre-trained model was further evaluated over four CRC-related tasks involving datasets from multiple countries. In summary, we proved that a multi-country self-supervised pre-trained model could effectively boost the performance of downstream validation datasets in various countries and be more robust in the external dataset.

2. Methodology

2.1. Overall pipeline

We designed a pipeline that enables the pre-training of AI models under SSL over a multi-country dataset, and achieves superior performance on four challenging downstream tasks in CRC (Figure 1). Although the ImageNet pre-trained model can be employed to extract features in a straightforward manner, there are significant morphological differences between pathological and natural images. In addition, limited by the lack of annotation in pathological images, self-supervision is considered a promising solution. To verify the superiority and robustness of SSL over other countries, we designed an SSL pre-training and downstream task verification framework for CRC. First, we built a multi-country CRC dataset from MCO (1611 WSIs from Australia) and TCGA-COAD/READ (637 WSIs from the United States) for pre-training. As shown in Figure 1a, all WSIs were first cut into a series of patches, and then we adopted ResNet-50 trained on NCT-CRC-HE to build a more informative pre-training dataset from TCGA and MCO. Moreover, based on the pre-training dataset, as shown in Figure 1b, a self-supervised method was used to pre-train the AI model. Specifically, we utilized SimSiam to pre-train ResNet-18. Finally, as illustrated in Figure 1c, the pre-trained model was evaluated on four multi-country downstream CRC tasks, including

fully supervised tissue classification, MSI vs. MSS classification, weakly supervised polyps types classification, adenoma grade classification and CRC grading, and five-year survival prediction.

2.2. Self-supervised methods for pre-training

To efficiently extract pathological image features, we adopted the state-of-the-art self-supervised pre-training method SimSiam. SimSiam (Chen and He, 2021) is a contrastive SSL method that learns image representations using Siamese networks. It takes as inputs x_1 and x_2 which are two randomly augmented views from an image x . An encoder network f consisting of a backbone (e.g., ResNet (He et al., 2016) with the last linear classification layer discarded) and a projection multi-layer perceptron (MLP) (Chen et al., 2020) is used to process x_1 and x_2 . The weights of encoder f are shared when processing the two views x_1 and x_2 . Subsequently, a prediction MLP (Grill et al., 2020) is used to transform the output of one view and match it to the other view by minimizing their negative cosine similarity. The details are provided in the Appendix.

2.3. Validation setup for downstream tasks

Considering that different epoch numbers of pre-training can affect the downstream task performance and that the AI model appears to have converged after being pre-trained for 20~40 epochs, we took six pre-trained models with different pre-training epochs (i.e., 20, 24, 28, 32, 36 and 40) for downstream tasks 1 and 2, and reported their average performance. We adopted two common evaluation protocols in our downstream validation experiments: fine-tuning and the KNN algorithm. The details are provided in the Appendix. Because the image size in tasks 3 and 4 is very large (10 million pixels), it is impractical to process images in a supervised manner. Therefore, we adopt the weakly supervised method, which is carried out in the following steps: 1) Cut the large image into a series of small patches; 2) use the pre-trained model to extract features from the small patches, and 3) train and test the performance of a series of weakly supervised models under different feature representations. The details are provided in the Appendix.

3. Datasets

3.1. Ethics statement.

The datasets used in this study were obtained with appropriate permission and ethics approval from the respective data custodians. The TCGA, NCT-CRC-HE, texture-CRC, TCGA-CRC-DX, PAIA2020, ECCG and UnitoPatho datasets were all publicly available. The MCO CRC dataset was approved by the UNSW Sydney Research Ethics Committee (UNSW HREAP 2014-7-49 and UNSW HREAP, HC17439). All patients with MCO CRC provided written informed consent.

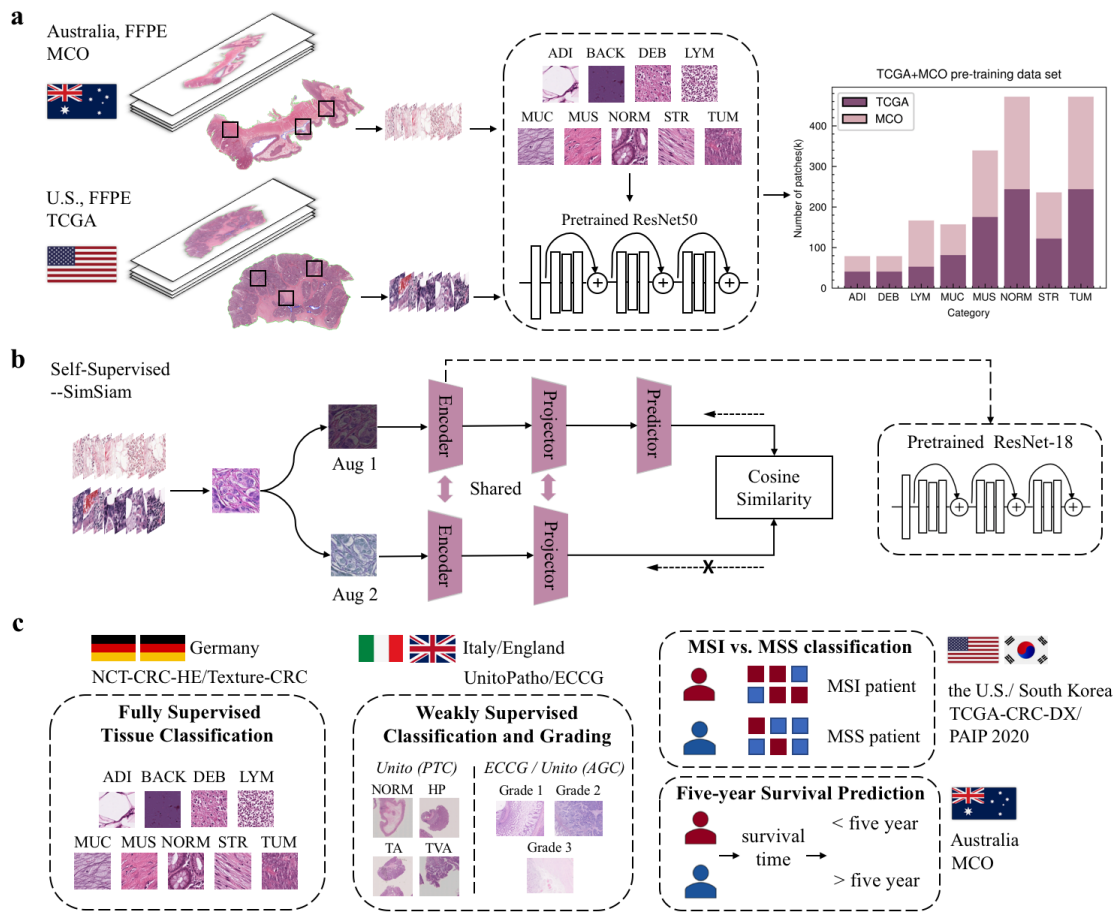


Fig. 1. Schematic of the deep learning and SSL workflows for CRC analysis. **a**, Multi-country dataset construction, a ResNet-50 is trained in NCT-CRC-HE dataset as a tissue classifier to select various patches from the MCO dataset (Australia) and TCGA dataset (the U.S.). Then TCGA pre-training dataset, MCO pre-training dataset, and TCGA+MCO pre-training dataset are used for self-supervised training. **b**, Models employed for self-supervised training, the pre-training method SimSiam is used to obtain the pre-trained ResNet-18 model. **c**, Downstream tasks validation, including fully supervised tissue classification, patient-level classification of microsatellite instability (MSI) vs. microsatellite stability (MSS), weakly supervised classification and grading, and five-year survival prediction.

Table 1. Datasets used from multi-center data sources

Data source	Dataset usage	Sample preparation	Population	Number
MCO	Model pre-training	FFPE	Australia	1,611 WSIs
TCGA-COADREAD	Five-year Survival Prediction			
NCT-CRC-HE	Model pre-training	FFPE	U.S.	638 WSIs
texture-CRC	Tissue classification	FFPE	Germany	107K patches
	Tissue classification (external testing)	FFPE	Germany	5K patches
TCGA-CRC-DX	MSI vs. MSS classification	FFPE	U.S.	192,312 patches
PAIP2020	MSI vs. MSS classification (external testing)	FFPE	Korea	47 WSIs
UNITOPATHO	Weakly-supervised colorectal polyps types classification /adenomas grading	FFPE	Italy	8669 patches (1812×1812)/ 867patches (15,855×15,855)
ECCG	Weakly-supervised cancer grading	FFPE	England	300 patches (4548×7548)

3.2. Pre-training datasets

We construct 3 large CRC histopathological datasets for pre-training the AI model: namely TCGA, MCO and Both (TCGA+MCO). We cut each WSI in above datasets into patches sized of 224×224 at $20\times$ magnification. After removing the background, the WSI in TCGA contains an average of 10,939 patches, a maximum of 51,354 patches and a minimum of 225 patches. For MCO, the WSI contains an average of 18,182 patches, a maximum of 72,896 patches and a minimum of 152 patches. Next, we use the whole NCT-CRC-HE dataset to train a tissue classifier (ResNet-50 (He et al., 2016)) for obtaining pseudo labels used in patch selection, and then employ selected patches to construct pre-training datasets. Specifically, the construction of the pre-training dataset needs to meet the following principles: 1) It is necessary to ensure that at least a certain number (500) of patches are taken from most WSI. 2) We intend to select more patches from the categories to which pathologists pay more attention by imposing a constraint on the ratios of different categories. Specifically, for MCO (TCGA), we select 38K (40K), 38K (40K), 114K (53K), 76K (81K), 164K (175K), 228K (244K), 114K (122K) and 228K (244K) patches for Adipose (ADI), background (BACK), debris (DEB), lymphocytes (LYM), mucus (MUC), smooth muscle (MUS), normal colon mucosa (NORM), cancer-associated stroma (STR) and colorectal adenocarcinoma epithelium (TUM), respectively. 3) For the construction of the TCGA+MCO dataset, the proportion of TCGA and MCO in the mixed dataset is 1:1. 4) The priority of patch selection in 1), 2) and 3) is determined by patch confidence score given by ResNet-50.

3.3. Downstream evaluation datasets

We evaluate the performance of the AI model over seven downstream evaluation datasets (two of them are used for external testing) which consist of patches sized of 224×224 at $20\times$ magnification and correspond to four tasks:

1) The NCT-CRC-HE and Texture-CRC datasets are employed for the tissue classification of CRC, where NCT-CRC-HE is used for internal testing and Texture-CRC is used for external testing. The NCT-CRC-HE dataset contains 107,180 patches from H&E stained WSIs comprising human CRC and normal tissue, in which 100,000 patches are split into training data and 7,180 image patches are split into testing data. The dataset covers nine tissue classes: Adipose (ADI), background (BACK), debris (DEB), lymphocytes (LYM), mucus (MUC), smooth muscle (MUS), normal colon mucosa (NORM), cancer-associated stroma (STR) and colorectal adenocarcinoma epithelium (TUM). We take 9,000 patches (1,000 random patches of each category) for fine-tuning and remain all 7,180 patches in testset for testing. We randomly sample 1,800 patches from the training set to form the validation set. The Texture-CRC dataset contains 5,000 images of formalin-fixed paraffin-embedded (FFPE) human colorectal adenocarcinomas (primary tumors) from the pathology archive at the Heidelberg University Medical Center. It contains eight classes of histology images adipose, debris, lymphoma, mucosa, stroma(simple),

stroma(complex), tumor, and empty. We discard all the images of mucosa category in Texture-CRC dataset and use it as an external testset. The label mapping relation between NCT-CRC-HE and Texture-CRC is given in the supplementary Table.

2) The TCGA-CRC-DX and PAIP2020 datasets are used for biomarker prediction, where TCGA-CRC-DX is used for internal testing and PAIP2020 is used for external testing. The TCGA-CRC-DX dataset contains 192,312 patches (93,408 in training set and 98,904 in testing set) derived from histological images of CRC patients in the TCGA cohort. All images in TCGA-CRC-DX dataset are derived from FFPE diagnostic slides ("DX" at the GDC data portal). We use the open-access version of this dataset from <https://zenodo.org/record/2530835#.Ymos201ByUk>. The PAIP2020 dataset comes from PAIP2020 grand challenge <https://paip2020.grand-challenge.org>, and we take the training data consisting of 47 WSIs as our external testset.

3) The UnitoPatho and ECCG are used for the CRC classification and grading, respectively. For UnitoPatho, we use 867 patches ($15,855 \times 15,855$ pixels) to carry out weakly supervised colorectal polyps types classification. Polyps types classification can be divided into: Normal tissue (NORM, 46 patches), Hyperplastic Polyp (HP, 45 patches), Tubular Adenoma (TA, 367 patches), Tubulo-Villous Adenoma (TVA, 161 patches). Adenomas grading task can be divided into: LG (TA-LG, TVA-LG, 414 patches), HG (TA-HG, TVA-HG, 114 patches), and None (Norm, HP, 91 patches). For ECCG, we use 300 patches ($4,548 \times 7,548$ pixels) to perform weakly supervised cancer grading. Each image is labelled as normal tissue (120 patches), low grade (120 patches) tumours or high grade (60 patches) tumours.

4) The MCO dataset and its follow-up data are used for the CRC five-year survival prediction. The MCO data includes 1,491 WSIs corresponding to 1,409 patients. We define data censored within five years as invalid data, and after filtering out this part of the data, the remaining 1,438 WSIs correspond to 1,359 patients. There are 571 WSIs for deaths within five years and 867 WSIs for non-death data within five years. Since a patient may have one or more WSIs, we consider that all WSIs of a patient inherit the same patient survival label.

It should be noted that for the weakly supervised downstream tasks including task 3 and 4, we adopt various commonly used weakly supervised models for training and testing, including LossAttn (Shi et al., 2020), TransMIL (Shao et al., 2021), AB-MIL (Ilse et al., 2018), and DSMIL (Li et al., 2021).

4. Experiments

4.1. SSL can improve fully supervised tissue classification for CRC

We employed SimSiam to pre-train the ResNet-18 model and then fine-tuned the pre-trained encoder for a downstream fully supervised tissue classification task. To compare ImageNet initialization and SSL initialization of the AI models, we employed two common evaluation protocols, namely, fine-tuning and the K-nearest-neighbor algorithm (KNN), to evaluate the performance over the NCT-CRC-HE dataset (overall 107,800

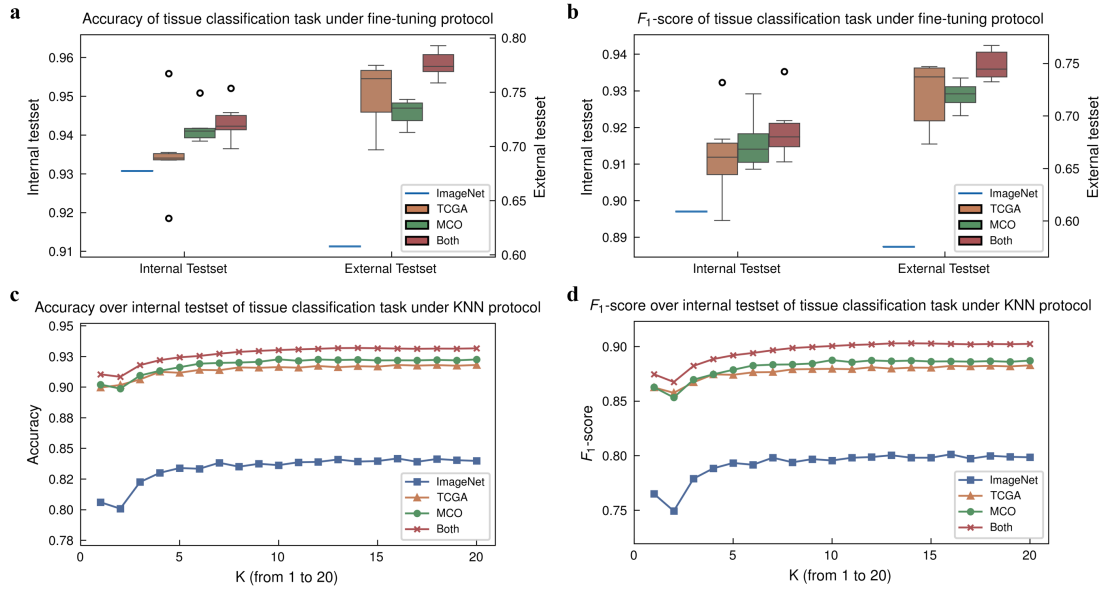


Fig. 2. Results of CRC tissue classification for ResNet-18 (pre-trained by SimSiam). **a**, Classification accuracy over internal and external testset under fine-tuning evaluation protocol. The internal testset refers to NCT-CRC-HE testset and the external testset refers to Texture-CRC dataset. **b**, Classification F_1 -score over internal and external testset under fine-tuning evaluation protocol. In **a** and **b**, the boxes show the median values and quartiles, the whiskers show the rest of the distribution (except for points identified as outliers which have a distance of 1.5 times the IQR from quartiles, where IQR is the distance between the upper and lower quartiles). **c**, Classification accuracy over internal testset under KNN evaluation protocol. **d**, Classification F_1 -score over internal testset under KNN evaluation protocol. In **c** and **d**, multiple KNN classifiers with K varying from 1 to 20 are constructed and the model performance curves with K are plotted.

patches from Germany; 9,000 patches are used for training and 7180 patches are used for testing, more details are presented in the Appendix). Meanwhile, we reported the performance over the texture-CRC dataset, which was used as an external test set (5,000 patches from Germany; 4,275 patches were used for testing) under the fine-tuning protocol.

Under the fine-tuning evaluation protocol, we found that ResNet-18 achieved an ACC/ F_1 of 0.9308/0.8971 over the internal testset (NCT-CRC-HE) and an ACC/ F_1 of 0.6080/0.5756 over the external testset (texture-CRC) when using ImageNet initialization (Figure 2 and Supplementary Tables 1 and 3). In comparison, self-supervised pre-training over large colorectal pathological datasets improved the performance of the AI model. Specifically, ACC/ F_1 was improved by 0.45%/1.51%, 1.12%/1.87%, and 1.26%/2.24% over the internal test set when SimSiam was used to pre-train the backbone over TCGA, MCO, and both(TCGA+MCO) datasets, respectively. The improvements over the external test set were even more significant, achieving 14.05%/14.51%, 12.34%/14.42%, and 16.78%/17.27%, respectively (Figure 2 and Supplementary Table 3). Under the KNN (integer K from 1 to 20 are tested) evaluation protocol, we evaluated the performance over the internal testset and found that ResNet-18 achieved the best ACC/ F_1 of 0.8418/0.8013 ($K=16$) when ImageNet initialization was used (Figure 2 and Supplementary Table 2). In comparison, self-supervised pre-training over large colorectal pathology datasets was found to significantly improve the performance of ResNet-18. Specifically, the best ACC/ F_1 was improved by 7.63% ($K=16$)/8.15% ($K=20$), 8.24%/9.34% ($K=14$), and 8.97%/10.07% ($K=20$) when SimSiam was used to pre-train

the backbone over TCGA, MCO, and both (TCGA+MCO) datasets, respectively (Figure 2 and Supplementary Table 2).

In summary, we observed that compared with fully supervised pre-training over ImageNet, the AI model could obtain better performance after self-supervised pre-training over large colorectal pathological datasets. In addition, pre-training over a multi-country dataset (TCGA+MCO) showed excellent performance compared with pre-training over a single-country dataset (TCGA or MCO) under both the fine-tuning protocol (ACC/ F_1 +0.81%/+0.73% against TCGA, ACC/ F_1 +0.14%/+0.37% against MCO over internal test set, ACC/ F_1 +2.73%/+2.76% against TCGA, ACC/ F_1 +4.44%/+2.85% against MCO over external test set) and the KNN protocol (ACC/ F_1 +1.34%/+1.92% against TCGA, ACC/ F_1 +0.73%/+0.73% against MCO over internal test set). The significant improvement of ImageNet initialization over the external test set indicates a much better generalization ability of the AI model with SSL initialization, which is desired in practical scenarios. Another interesting observation is that pre-training over the TCGA dataset led to better generalization over the external test set than pre-training over the MCO dataset, indicating that pre-training over a dataset consisting of images with more variation can greatly promote the generalization of the model.

4.2. SSL can improve fully supervised biomarker prediction for CRC

Similar to the process in section 4.1, we transferred the pre-trained backbone of ResNet-18 to another downstream binary classification task (MSI vs. MSS prediction). Considering that the patient-level classification performance highly

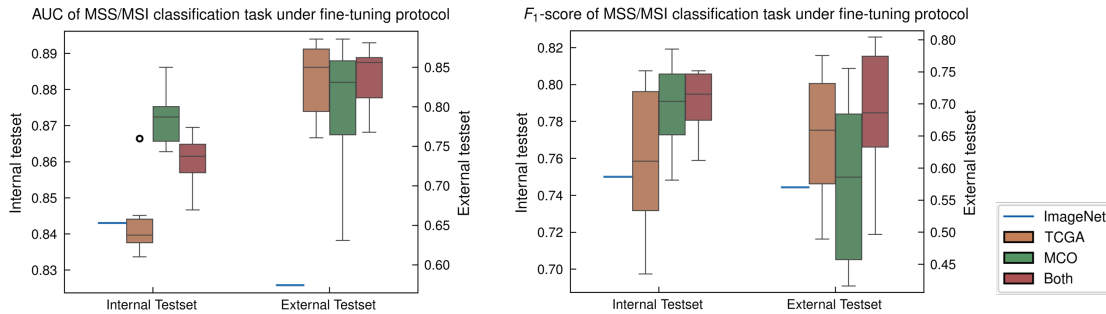


Fig. 3. Results of MSI prediction with ResNet-18 (pre-trained by SimSiam). **a**, Classification AUC over internal and external testset under fine-tuning protocol. The internal testset refers to the TCGA-CRC-DX dataset and the external testset refers to the PAIP2020 dataset. **b**, Classification F_1 -score over internal and external testset under fine-tuning protocol. In **a** and **b**, the boxes show the median values and quartiles, the whiskers show the rest of the distribution (except for points identified as outliers which have a distance of 1.5 times the IQR from quartiles, where IQR is the distance between the upper and lower quartiles).

depends on the patch-level classification performance in this task, and the relatively poor performance under the KNN protocol may lead to unreasonable patient-level results, we only adopted the fine-tuning protocol to train a patch-level classifier whose predictions for patches were used for acquiring patient-level prediction. We evaluated the patient-level performance of macro areas under the receiver operating curve (AUC) and F_1 -score for the prediction of MSI over the TCGA-CRC-DX cohort (387 patients from TCGA; more details of this dataset are presented in supplementary). We also evaluated the external testing performance of the PAIP2020 dataset. Under the fine-tuning evaluation protocol, we found that ResNet-18 achieved AUC/ F_1 of 0.8452/0.7376 over the internal testset (TCGA-CRC-DX) and AUC/ F_1 of 0.7881/0.5696 over the external testset (PAIP2020) when using the default ImageNet initialization (Figure 3 and Supplementary Tables 4 and 5). Compared with ImageNet initialization, self-supervised pre-training over large colorectal histopathology datasets improved the performance of the AI model. Specifically, AUC/ F_1 over the internal test set was improved by -0.16%/2.13%, 2.7%/5.02%, and 1.49%/5.26%, respectively, and the AUC/ F_1 over the external test set was improved by 4.56%/7.82%, 0.99%/0.94%, and 4.88%/11.24% when SimSiam was used to pre-train the backbone over TCGA, MCO, and both (TCGA+MCO) datasets, respectively (Figure 3 and Supplementary Tables 4 and 5). In summary, we can see that the model extracts better features and achieves higher performance on downstream MSI vs. MSS bio-marker prediction tasks after being pre-trained over large CRC datasets in a self-supervised manner. Meanwhile, pre-training over a multi-country dataset (TCGA+MCO) showed excellent performance compared with pre-training over a single-country dataset (TCGA or MCO) under the fine-tuning protocol (AUC/ F_1 +1.65%/+3.13% against TCGA and F_1 +0.24% against MCO over the internal testset, AUC/ F_1 +0.32%/+3.42% against TCGA and AUC/ F_1 +3.89%/+10.3% against MCO over the external testset). Moreover, pre-training over the TCGA dataset made the AI model generalize better to the external test set than pre-training over the MCO dataset, similar to the observation in section 4.1.

4.3. SSL can improve weakly supervised classification and grading for CRC

We evaluated weakly supervised CRC classification and grading in the UnitoPatho cohort (867 WSIs from Italy) and the ECCG cohort (300 WSIs from the United Kingdom), and the results are shown in Figure 4 and Supplementary Table 6. In weakly supervised tasks, a WSI often corresponds to only one label. Due to limitations in computing power, WSIs cannot be processed directly. Therefore, we adopted a general process: 1) splitting small patches, 2) extracting features using a pre-trained model, and 3) training a weakly supervised model. For the ECCG cohort, the self-supervised pre-trained models obtained from pathological images always showed a better performance than ImageNet. The mean metric gains of the SimSiam training were AUC and ACC of +0.41%, +1.51%; +0.67%, +2.16%; and +0.91%, +2.84% for TCGA, MCO, and TCGA+MCO, respectively. It can be seen that when using the multi-country dataset pre-training over the ECCG dataset, the downstream weakly supervised classification task can achieve better performance. Similarly, we also performed four-category polyp type classification (PTC) and three-category adenoma grade classification (AGC) using UnitoPatho. Among the mean metric gains of SimSiam training for UnitoPatho PTC, the AUC and ACC of TCGA, MCO, and TCGA+MCO were +2.24%, +3.55%; +0.95%, -0.20%; and +1.65%, +2.12%, respectively, and for UnitoPatho AGC, the AUC and ACC were +1.78%, +2.65%; +3.61%, +6.85%; and +4.41%, +6.57%, respectively. Overall, it can be seen that multi-country training still shows stable excellence.

4.4. SSL can improve the weakly supervised five-year survival prediction for CRC

We tested our SSL pipeline in another downstream task: CRC five-year survival prediction classification (1611 WSIs from the MCO cohort), and the results are shown in Figure 5 and Supplementary Table 7. After removing the data without follow-up time (details in Supplementary section 2.2), the MCO cohort retained 1438 WSIs corresponding to 1359 patients. We first cut the WSIs into a series of small patches and then utilized the pre-trained model to extract the features of the patches. Finally, we compared the results of four weakly supervised mod-

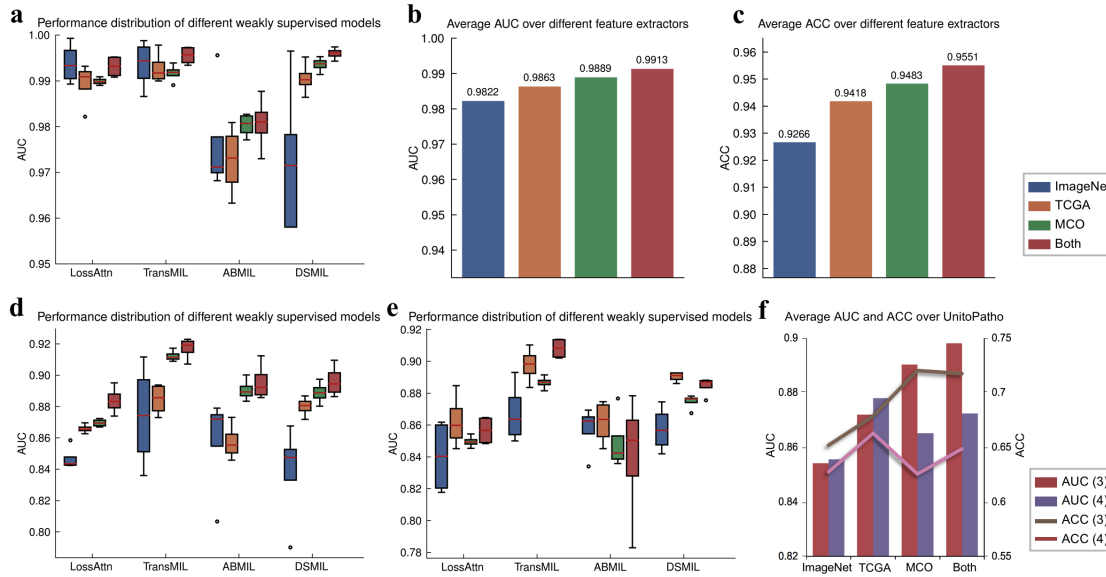


Fig. 4. Results of weakly supervised CRC classification and grading. **a, d, e,** performance distribution for different weakly supervised models over the ECCG grading task, UnitoPatho three-category AGC task and four-category PTC task, respectively. **b, c,** average AUC/ACC with different feature extractors (ImageNet/SimSiam pre-trained) over ECCG grading task. **f,** average AUC and ACC with different feature extractors (ImageNet/SimSiam pre-trained) over UnitoPatho three-category AGC task and four-category PTC task. In the boxplot, the x-axis represents the performance of different weakly supervised models on different pre-training features, and the y-axis represents the AUC interval of the same model on different data partitions. Boxes represent the 1st, median, and 3rd quartiles of the data distribution, with whiskers extending to data points within 1.5 times the interquartile range, and black circles represent outliers. In the bar chart, the four columns represent the average AUC performance of different weakly supervised models on the SimSiam training features. In the combination chart, AUC is shown in the form of a column chart, corresponding to the main coordinate axis. ACC is shown in the form of a line graph, corresponding to the secondary axis.

els (Shi et al., 2020; Shao et al., 2021; Ilse et al., 2018; Li et al., 2021) that used pre-trained features on different datasets. In the SimSiam pre-training manner, compared to ImageNet pre-trained feature results, the performance improved on average over TCGA (AUC, +0.25%, ACC, +0.80%), MCO (AUC, +1.60%, ACC, +1.67%), and TCGA+MCO (AUC, +1.35%, ACC, +1.57%). It can be seen from the results that for this task, the pre-training of the MCO dataset and the multi-country dataset can both achieve relatively better performance. This is mainly due to two reasons. First, the MCO dataset belongs to a specific CRC subtype; that is, the feature distribution of the dataset is relatively simple. Second, in this task, we noticed that the similarity degree between the MCO pre-training dataset and the five-year survival prediction downstream dataset is high, which makes the pre-training over the MCO dataset more advantageous. Nevertheless, multi-country training still has good adaptability and represents a higher application value.

5. Discussion

With the development of AI and increasing attention on health, computational pathology has become an important research direction globally. At the same time, with the continuous upgrading of digital pathology scanning technology, pathological diagnoses in hospitals worldwide have gradually tended toward digital processes, resulting in a large amount of data. However, unlike natural and medical images, such as computed tomography (CT) and magnetic resonance imaging (MRI), the high resolution and wide field of view of pathological images

make image annotation a huge challenge. Therefore, pathological images often lack fine local annotations or even have no annotations. Current AI methods usually take weights pre-trained on the natural image dataset, that is, ImageNet, to initialize the parameters rather than using pathological images under fully supervised pre-training. Fortunately, SSL, as a common method of using unlabeled data, can learn the features of the image itself by designing pretext tasks. Therefore, a self-supervised model based on pathological images is ideal. However, there is significant variance among the data collected in different countries, which is mainly attributed to the following reasons: 1) different physiological characteristics of patients in different countries and regions, 2) differentiation of pathological slide preparation in different countries and regions, and 3) differentiation of diagnostic criteria in different countries and regions. Therefore, a self-supervised model trained in only one country cannot achieve satisfactory results for the generalization of multi-country data. To address this problem, we conducted a series of explorations for multi-country data training and testing of CRC.

As a common malignant cancer worldwide, research on CRC is clinically significant. Typical tasks in CRC today include the analysis of tissue types, grading of cancer growth patterns, classification of the polyps malignancy degree, analysis of biomarker expression, and study of prognostic treatments. For tissue type classification, decomposing complex tissue into different tissue categories can help doctors to better analyze the micro environment information of tumors (Kather et al., 2019a). The grading task helps doctors judge the tumor's current growth pattern and predict its growth and spread according

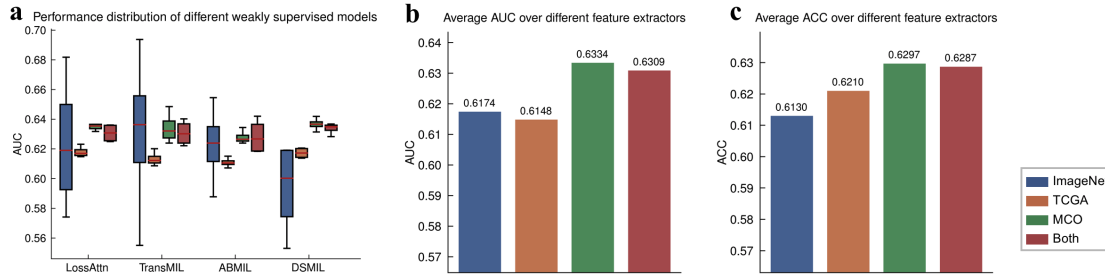


Fig. 5. Results of CRC five-year survival prediction. a, results of performance distribution for different weakly supervised models over the MCO dataset. **b,** **c,** average AUC and ACC for different features over the MCO dataset. In the boxplot, the x-axis represents the performance of different weakly supervised models on different pre-training features, and the y-axis represents the AUC interval of the same model on different data partitions. Boxes represent the 1st, median, and 3rd quartiles of the data distribution, with whiskers extending to data points within 1.5 times the interquartile range, and black circles represent outliers. In the bar chart, the four columns represent the average AUC performance of different weakly supervised models on the SimSiam training features.

to the tumor's degree of malignancy (Barbano et al., 2021). Examination of colorectal polyps can help doctors identify precancerous lesions in the intestinal mucosa and help patients avoid and predict invasive cancer by classifying the degree of polyp malignancy and dysplasia (Barbano et al., 2021). Biomarker analysis not only helps to elucidate the pathogenic mechanism of cancer but also helps doctors design treatment plans in a targeted manner. For example, MSI determines whether gastrointestinal cancer patients with gastrointestinal cancer respond particularly well to immunotherapy (Kather et al., 2019b; Echle et al., 2020). In addition, prognostic analysis of colon cancer can help doctors choose better treatment options, and retrospective prognostic studies can explore high-risk factors in cancer prognosis. For all the downstream tasks of CRC reported in this paper, including tissue classification, biomarker prediction, slide classification and grading, and five-year survival prediction, it can be seen that self-supervised pre-training in multiple countries can generally achieve better results. In the future, our pre-trained model on a large-scale CRC dataset could also be more widely applied to other downstream datasets, effectively promoting the level of AI medical care. To intuitively understand and compare the quality of representations extracted by different models, we adopted the uniform manifold approximation and projection (UMAP) technique (McInnes et al., 2018) to reduce the representations dimensionality for visualization. Supplementary Figure 1 shows the 2D UMAP projection of the patch representations (512 dimensions) extracted using ResNet-18. We observed that the patch representations extracted by ResNet-18 using SimSiam pre-training initialization tended to form more discriminative clusters across different categories; that is, representations of the same category tended to cluster together, while representations of different categories tended to be more distant. In addition, the clusters appeared to be more compact when using SimSiam both (lower right) pre-training initialization compared with using SimSiam TCGA/MCO (upper right/lower left) pre-training initialization, as shown in the red and purple rectangular boxes, which corresponds to the BACK and TUM categories, respectively. The UMAP projection result indicates that SimSiam pre-training helped the AI model learn patch representations that can better preserve the global structure of the original data and that SimSiam pre-training over

multi-country data was able to further promote the quality of patch representation compared to single-country data.

The pre-training datasets used in this study included the MCO dataset from Australia and the TCGA dataset from the United States. The spatial distribution of the dataset does not cover other areas, such as South America and Africa. In addition, in the downstream task test of this study, fully supervised classification and weakly supervised classification were tested on four important CRC tasks. Additional tasks, such as cell detection and gland segmentation, require further testing. Furthermore, we found that the construction of multi-country datasets is crucial for SSL in CRC. Here, we have proposed a method to assist the construction of a multi-country dataset based on an external nine-category dataset of CRC tissues and screen patches based on the importance of tissue categories and the complexity of their morphology. The construction method of this multi-country dataset is not ideal and requires further improvement. In the future, we plan to build a multi-country CRC dataset covering all continents based on a more sophisticated pre-training set construction mechanism. Furthermore, considering the superior performance of this method in CRC, we will apply it to more cancer categories.

6. Conclusion

CRC is a widespread and heterogeneous cancer that has important research value. In this study, we explored SSL on multi-country datasets for CRC. Specifically, this study includes self-supervised pre-training over two large CRC datasets (TCGA dataset from the United States and MCO dataset from Australia) and the evaluation of four common downstream tasks in CRC (fully supervised tissue classification, MSI vs. MSS classification, weakly supervised PTC, AGC and CRC grading, and five-year survival prediction). To demonstrate the generalizability of the multi-country training method, downstream task datasets were also obtained from multiple countries, including Germany, Italy, England, South Korea, the United States, and Australia. We conducted extensive and rigorous experiments to demonstrate that multi-country self-supervised training has superior performance and better generalization.

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Supplementary Material

Experimental results for all downstream tasks are tabulated in Supplementary Tables 1–9. Details about the method and experiments are described further in the Supplementary.

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